

Chapter C3: Chemicals of the natural environment

Overview

Our way of life depends on a wide range of products made from natural resources. The Earth's crust provides us with metal ores and crude oil and our use of these impacts on the natural environment. Chemistry is fundamental to an understanding of the scale and significance of this human activity.

In Topic C3.1 the properties of metals are related to their structure and bonding, and in Topic C3.2 learners discover why the reactivity of a metal determines how it is extracted from its ores and

how new technologies enable us to extract metals from poor quality ores.

Electrolysis is explained in Topic C3.3, and learners learn about the wide variety of products made by electrolysis.

Finally, Topic C3.4 covers the separation of crude oil into fractions and the use of these fractions to make other chemicals and polymers. Within this context learners study the properties of simple molecules in relation to covalent bonding and intermolecular forces.

Learning about chemicals of the natural environment before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- know the differences between atoms, elements and compounds
- be familiar with the use of chemical symbols and formulae for elements and compounds
- be familiar with the use of formulae and equations to represent chemical reactions
- understand chemical reactions as the rearrangement of atoms
- know about reactions of acids with metals to produce a salt plus hydrogen
- know some displacement reactions
- know the order of metals and carbon in the reactivity series
- know that carbon is used to obtain metals from metal oxides.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

Learning about chemicals of the natural environment at GCSE (9–1)

C3.1 How are the atoms held together in a metal?

Teaching and learning narrative

Chemists use a model of metal structure to explain the properties of metals (1aS3). In the model, metal atoms are arranged closely together in a giant structure, held together by attraction between the positively charged atoms and a 'sea' of negatively charged electrons. Metals are malleable and ductile because the ions can slide over each other but still be held together by the electrons; they conduct electricity and heat because their electrons are free to move; and they have high boiling points and melting points due to the strong electrostatic attraction between metal ions and the electrons. These properties of metals make them useful.

Assessable learning outcomes

Learners will be required to:

1. describe the nature and arrangement of chemical bonds in metals
2. explain how the bulk properties of metals are related to the type of bonds they contain

Linked learning opportunities

Ideas about Science:

- use the model of metal structure to explain properties of metals (1aS3)